

ORGANIC CHEMISTRY

organic chemistry

21/08/22

Naming an organic compound::

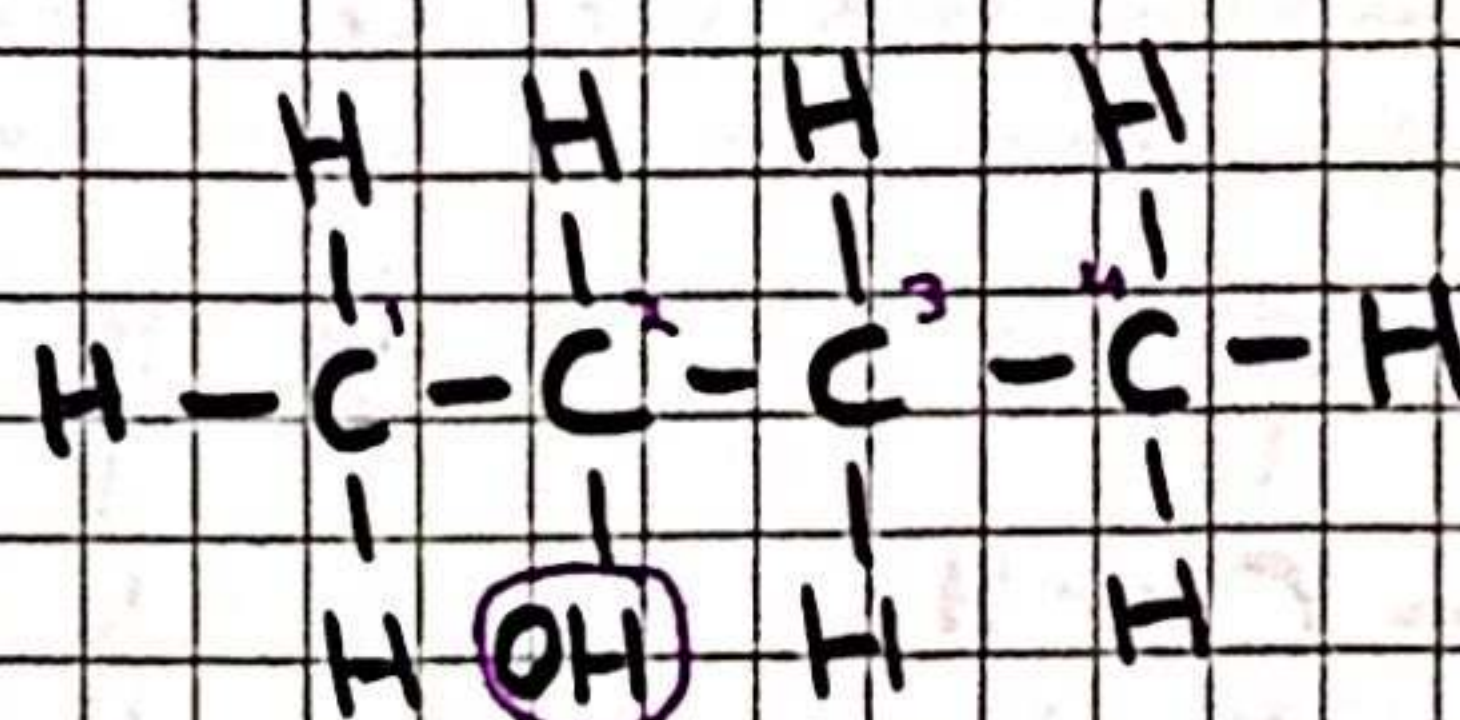
- find the prefix
- find the suffix
- find the position of the functional group

[prefix]

[suffix]

[position]

prefix	Carbon number	Compound	suffix
Meth	1	alkane	'—ane'
Eth	2		
Prop	3	alkene	'—ene'
But	4		
Pent	5	alcohol	'—anol'
Hex	6		
Hept	7	carboxylic acid	'—anoic acid'
Oct	8		
Non	9	ester	'—yl'
Dec	10		'—anoate'

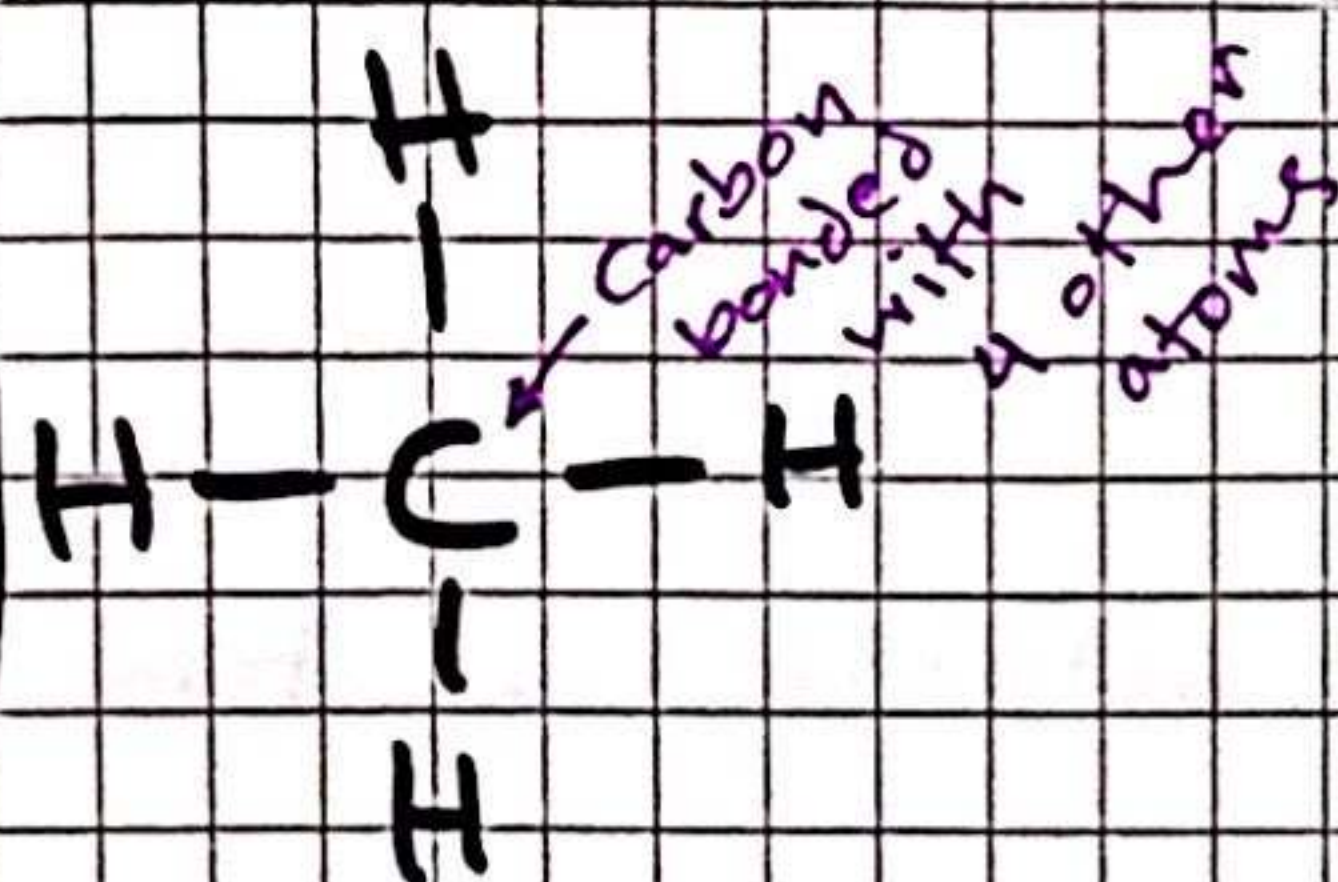


Butanol (alcohol)

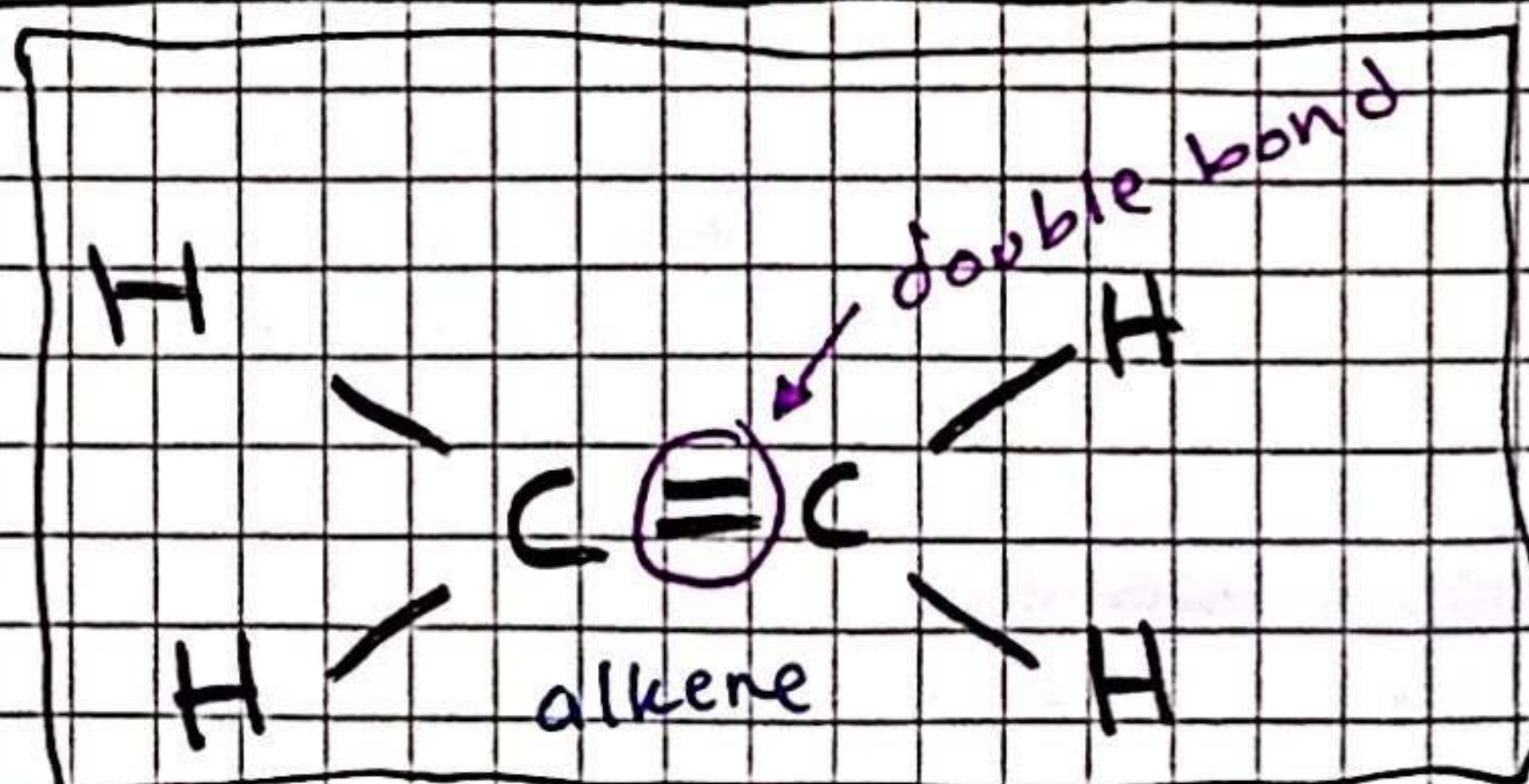
- the 'OH' is bonded with the 2nd carbon therefore it is:

2-Butanol

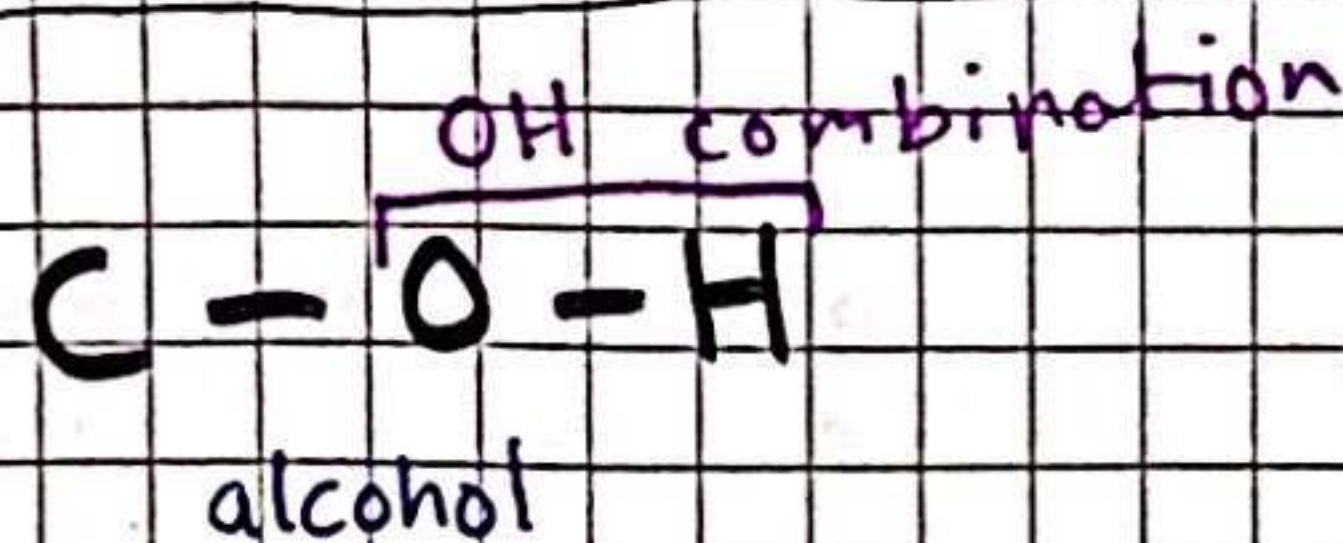
STRUCTURE::



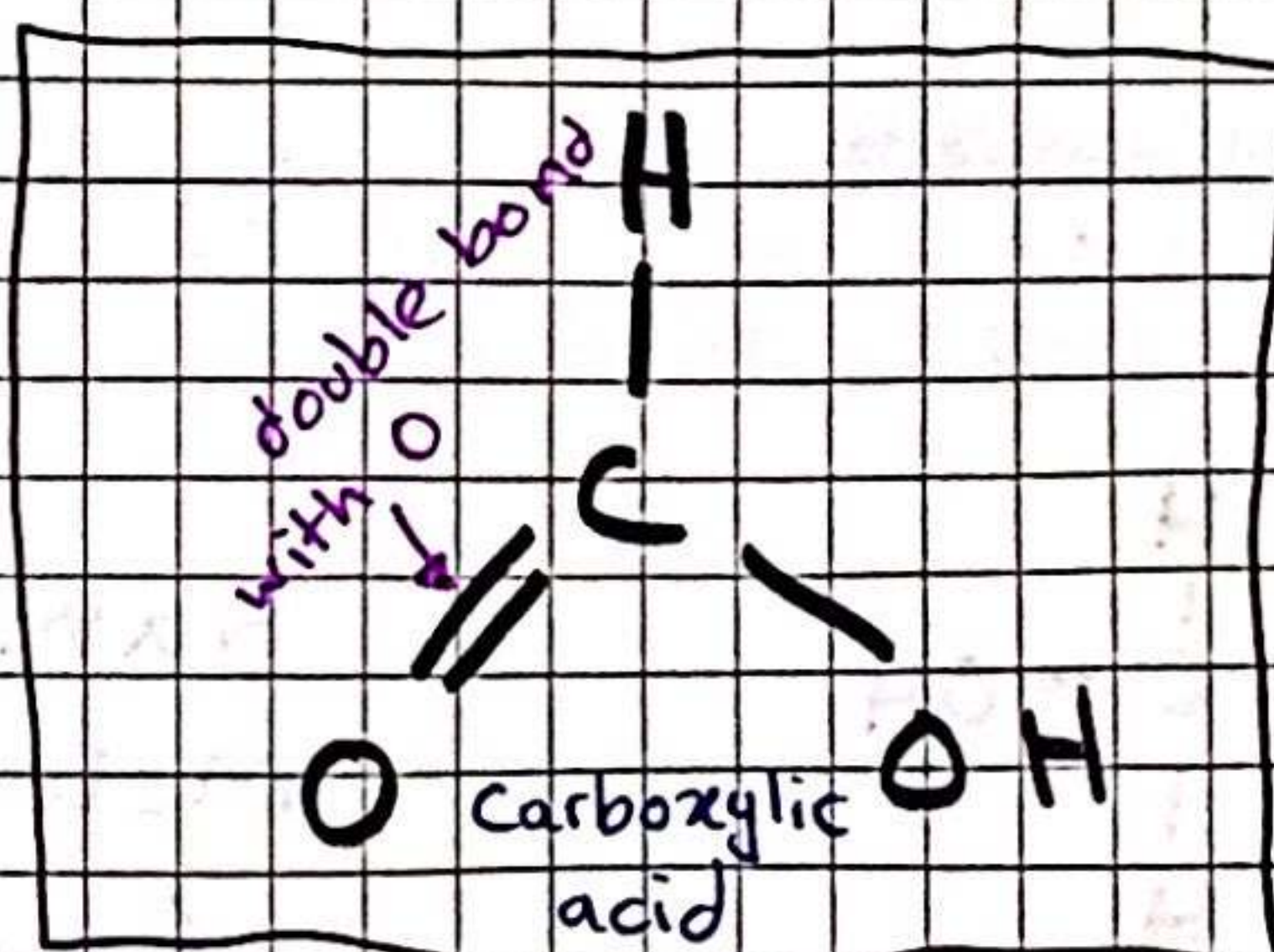
alkane



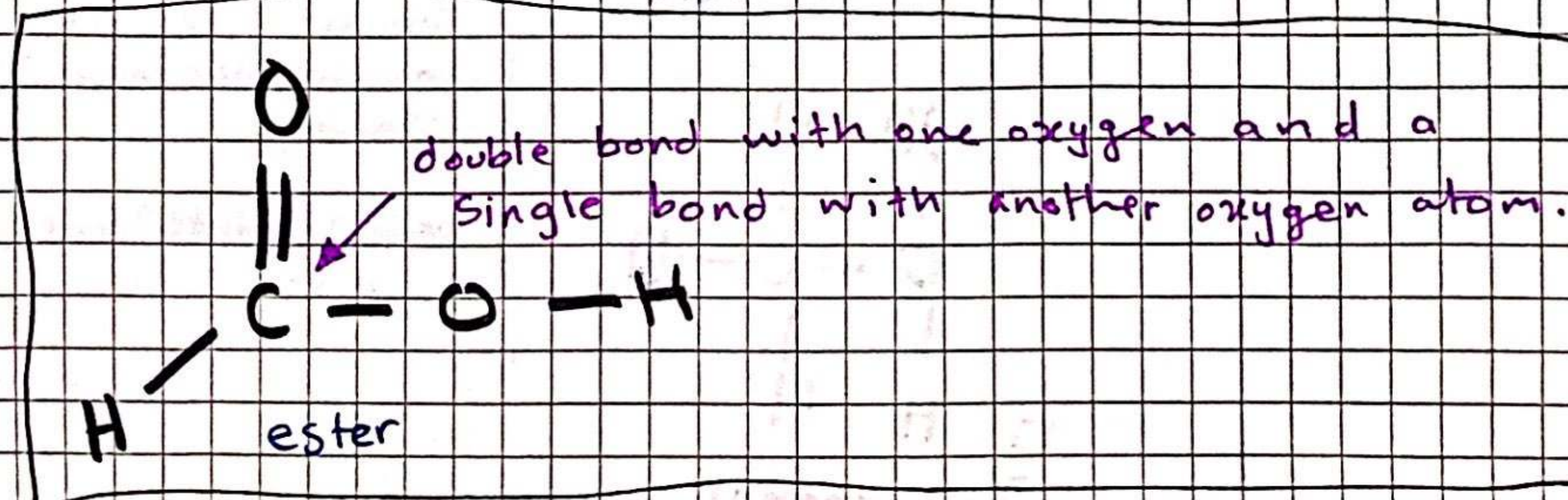
alkene



alcohol



Carboxylic acid

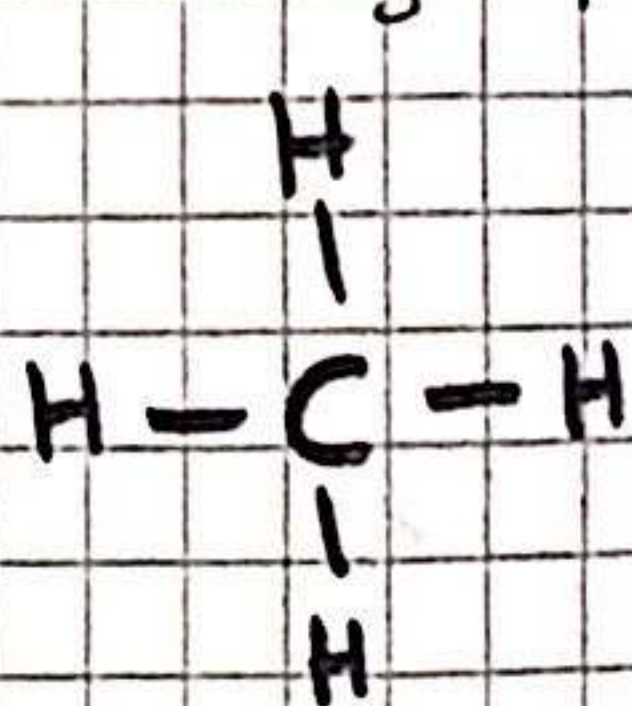


ester

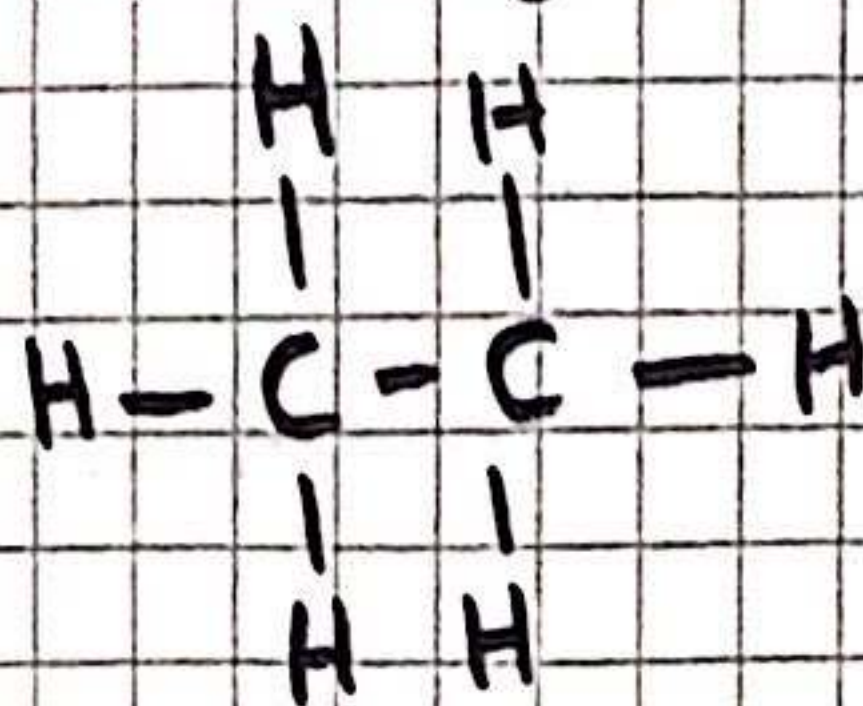
formulas

- alkane $\rightarrow C_n H_{2n+2}$
- alkene $\rightarrow C_n H_{2n}$
- alcohol $\rightarrow C_n H_{2n+1} OH$
- carboxylic acid $\rightarrow C_n H_{2n+1} COOH$

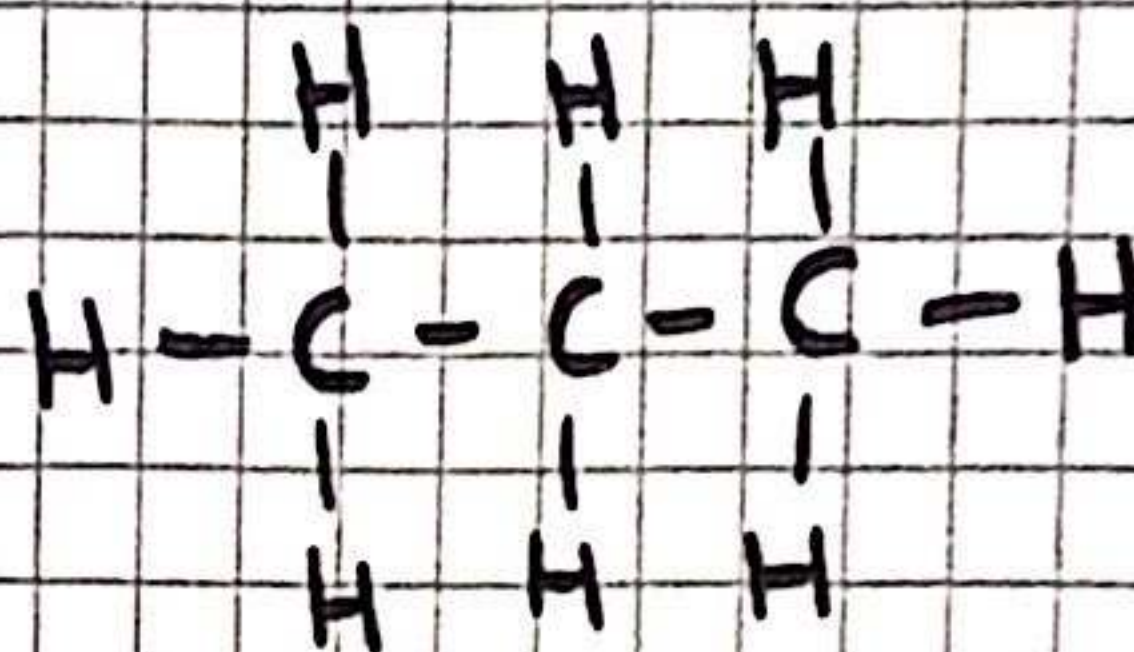
- homologous have similar chemical properties because their function group doesn't change.



methane



ethane



propane

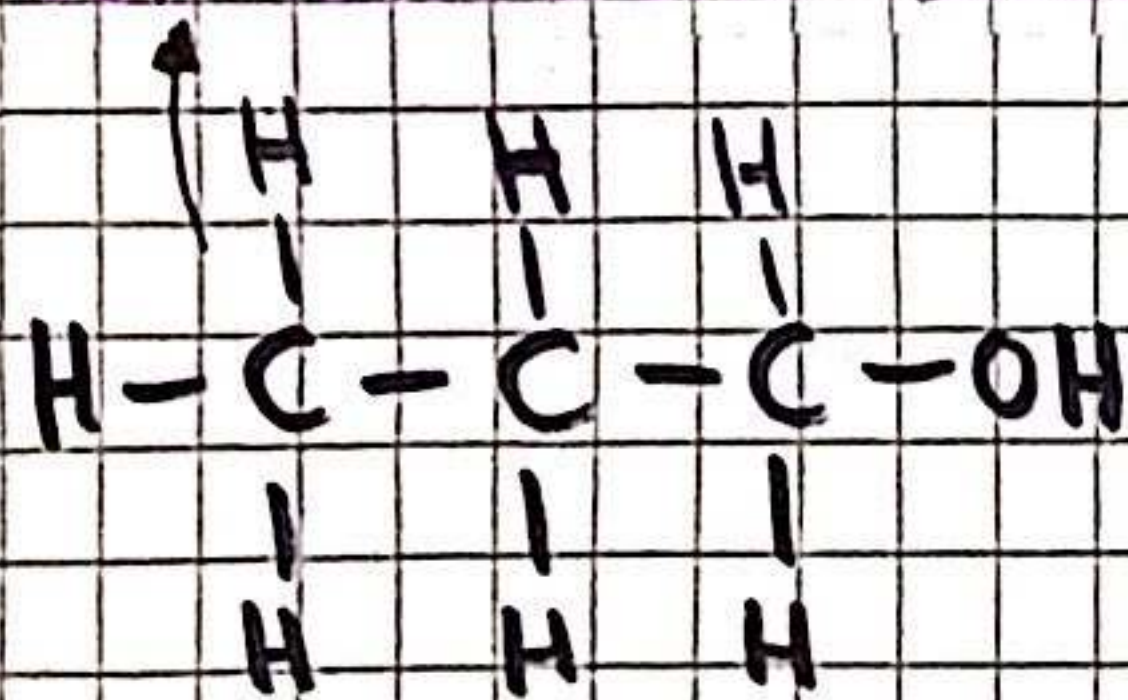
all three
only have one difference;
the number of CH_2
groups

structure formula:

to identify the location of chemical bonds between the atoms of a molecule.

Structural isomer:

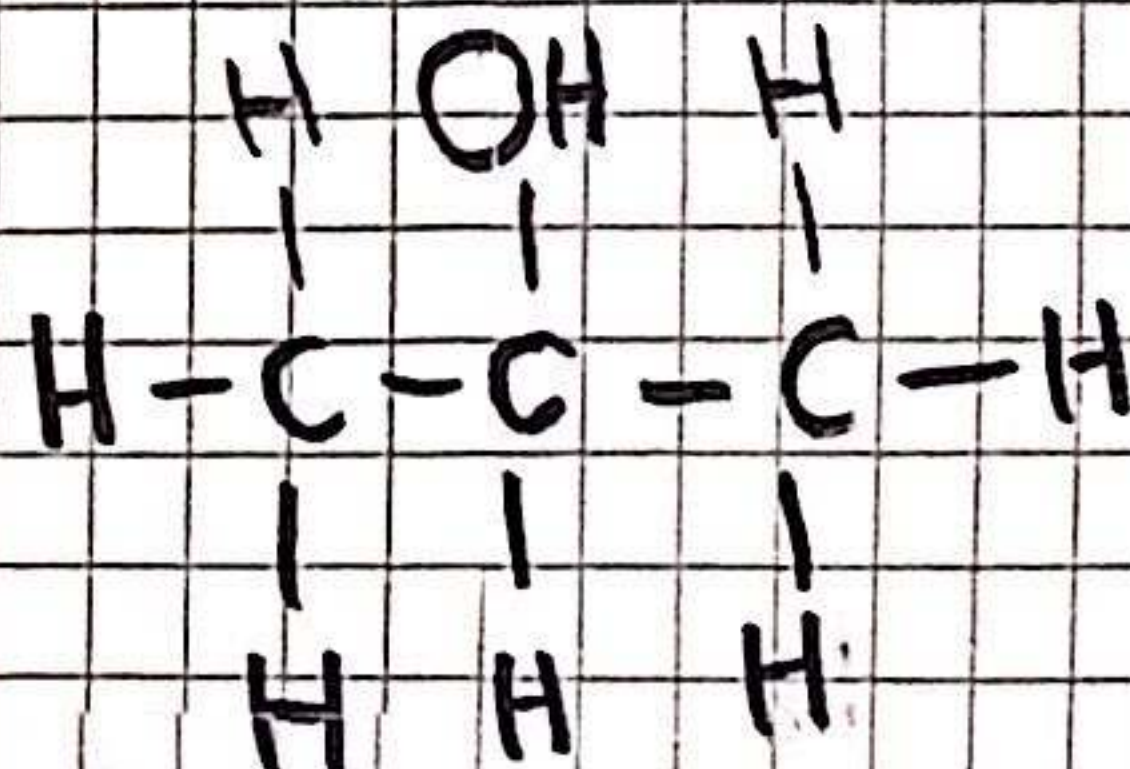
where two or more compounds have the same molecular formula but different structures.



1-propanol

Same
molecular
formula

different
structures



2-propanol

a saturated compound:
a compound with
molecules which
are all carbon-carbon
single bonds.
This kind neither has a
positive nor negative
charge.

an unsaturated compound:
a compound with
molecules in which one
or more carbon-carbon
bonds are not single
bonds.

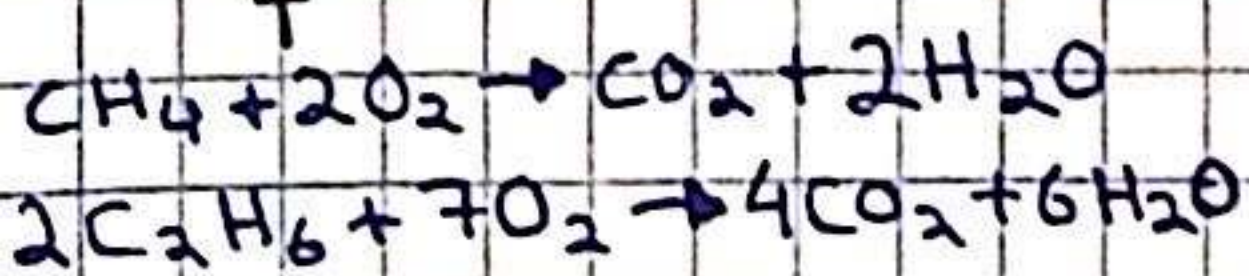
homologous series
a group of organic
compounds that
differ from one
another by $1 CH_2$

Alkanes:

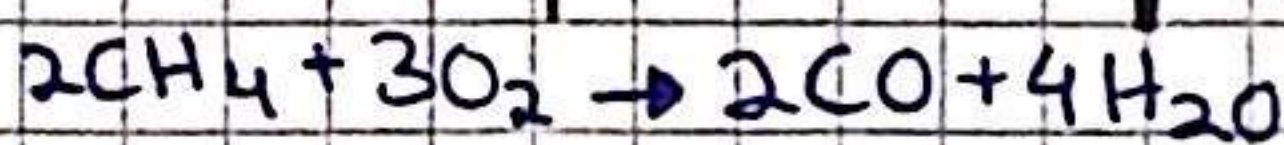
- hydrocarbons
- general formula $\rightarrow C_nH_{2n+2}$
- no more atoms can be added; saturated hydrocarbon
- ends with 'ane'
- unreactive except for combustion and chlorination

reaction between
an organic molecule
and oxygen in
the air.

complete combustion
is when there is
sufficient oxygen
to react with all
of the hydrocarbon,
and will give you
carbon dioxide +
water



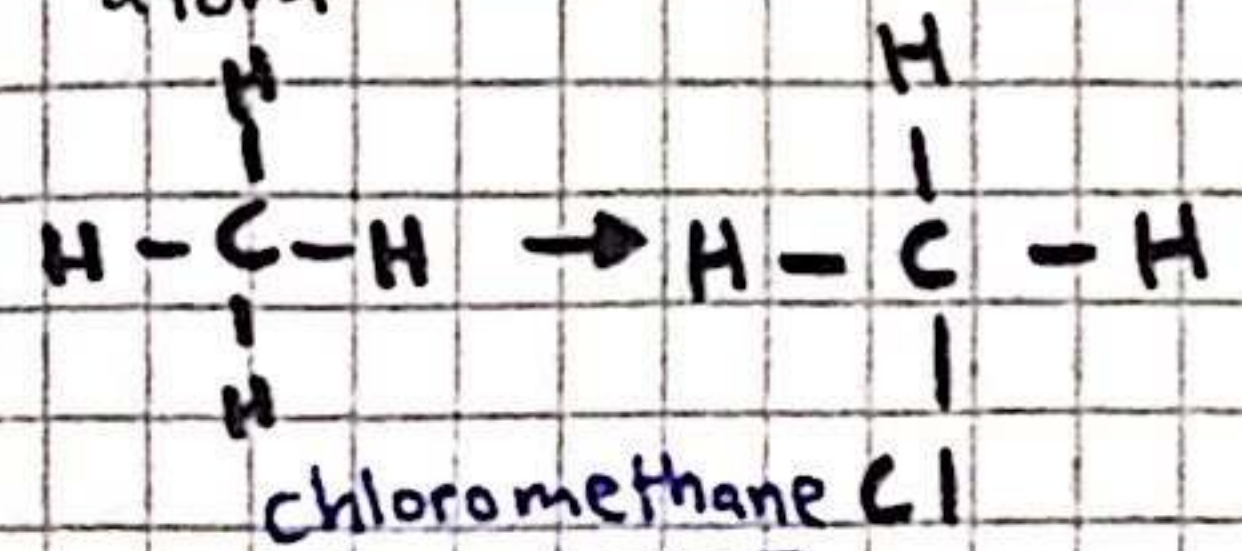
incomplete combustion
is when there is
insufficient oxygen
to react with all
of the hydrocarbon
so instead it
produces carbon
monoxide + water



Substitution

- alkane reacts with
chlorine in bright light
to give a mixture of
chloroalkanes

- one hydrogen atom is
replaced by one chlorine
atom



- if there is enough chlorine
then all the hydrogen atoms
get replaced.

Alkenes:

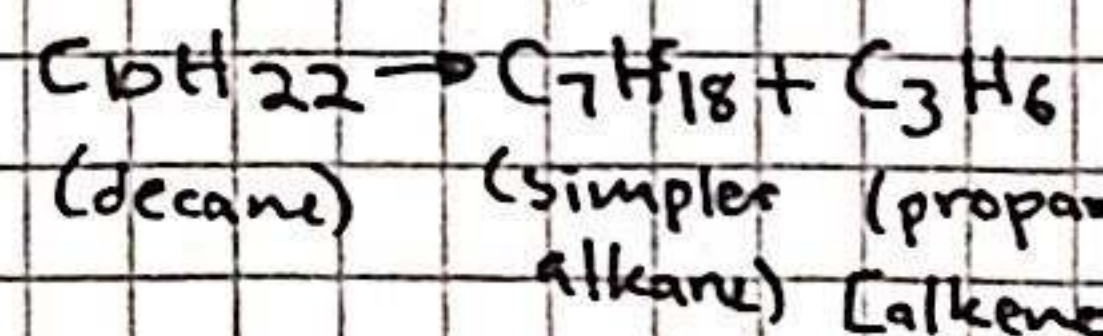
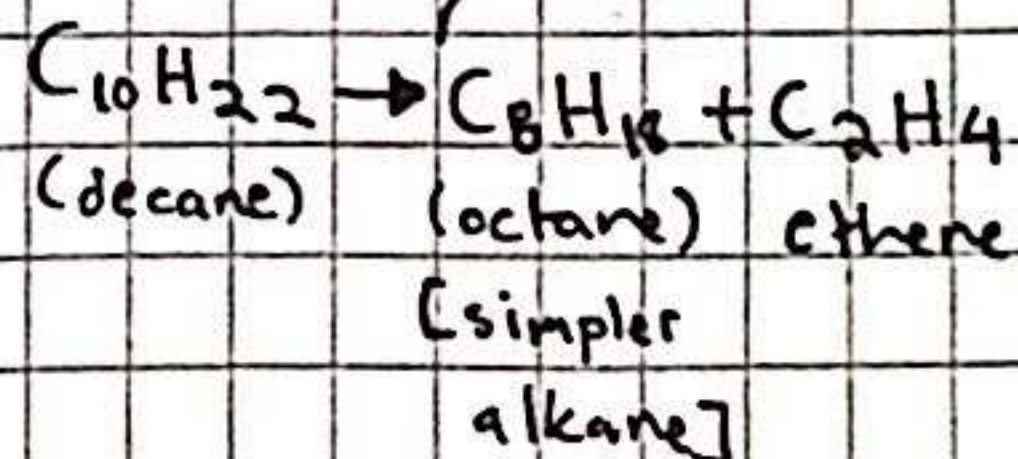
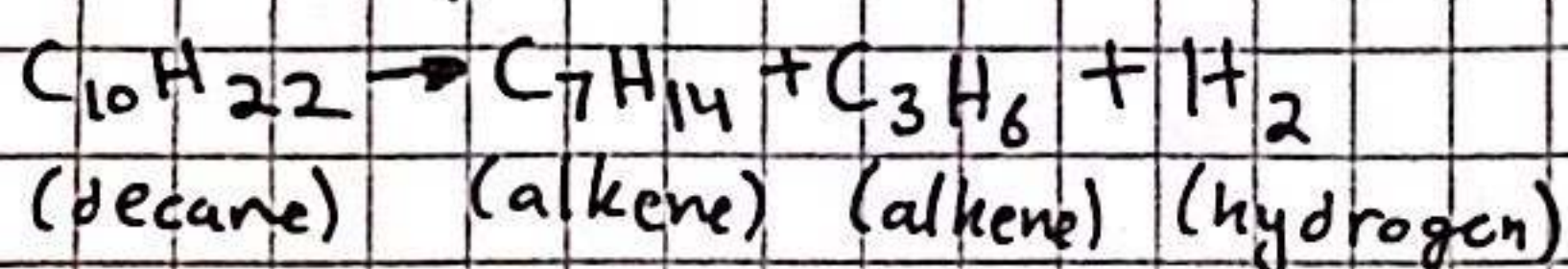
- hydrocarbons
- general formula $\rightarrow C_nH_{2n}$
- can undergo additional reactions therefore is an unsaturated hydrocarbon
- contains double bonds ($C=C$)
- ends with 'ene'

made of cracking alkanes

large alkane molecules are passed over a heated catalyst.

larger alkanes are broken down to simpler alkanes, alkenes
and simple hydrogen.

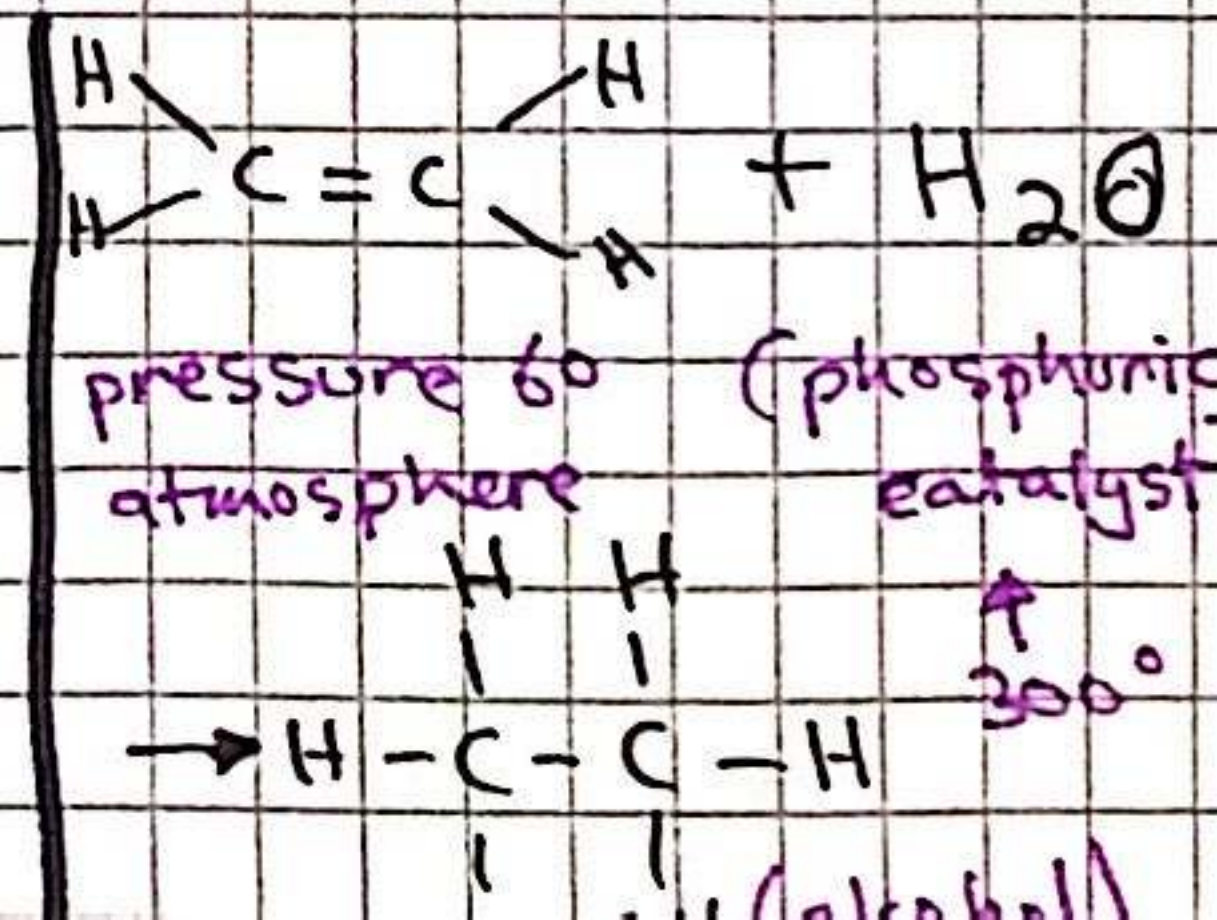
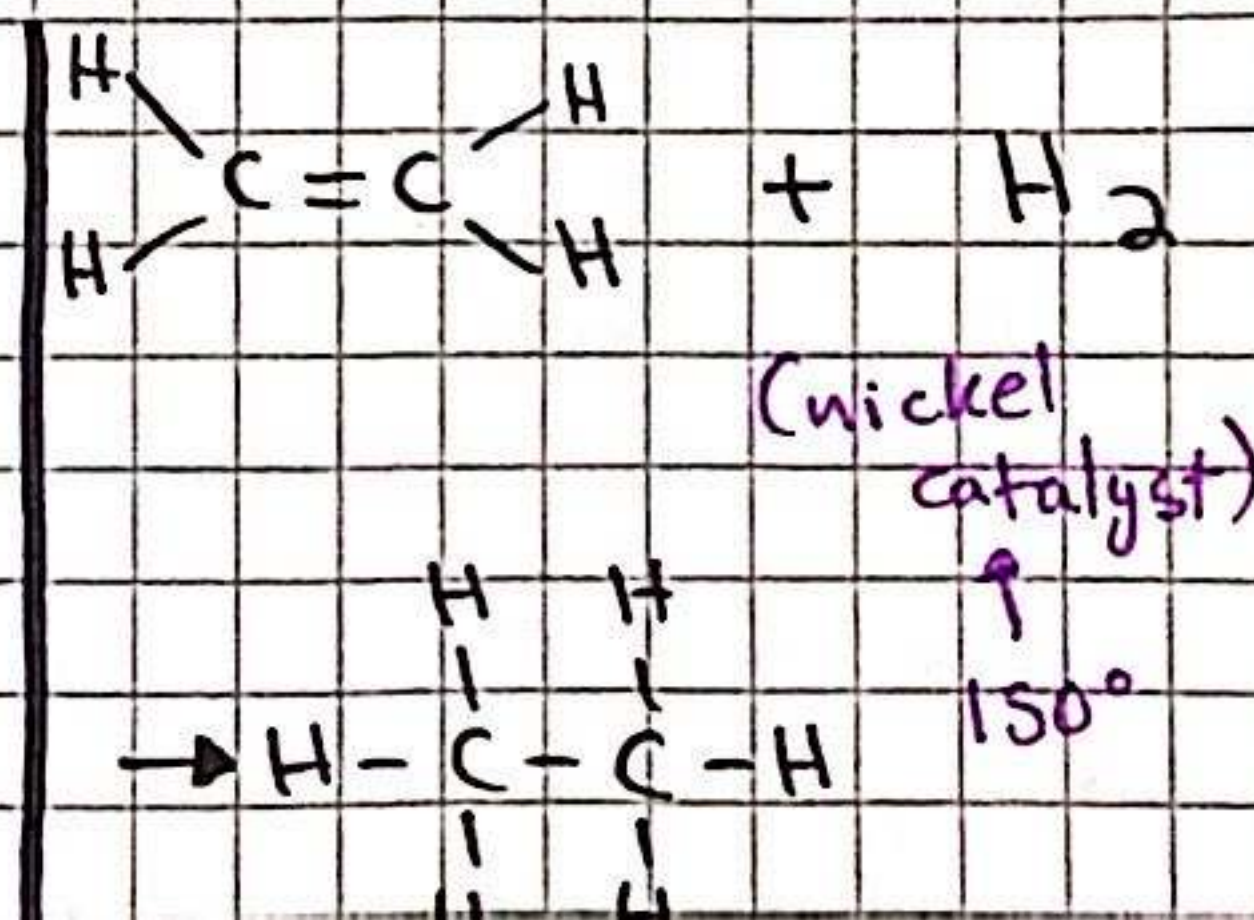
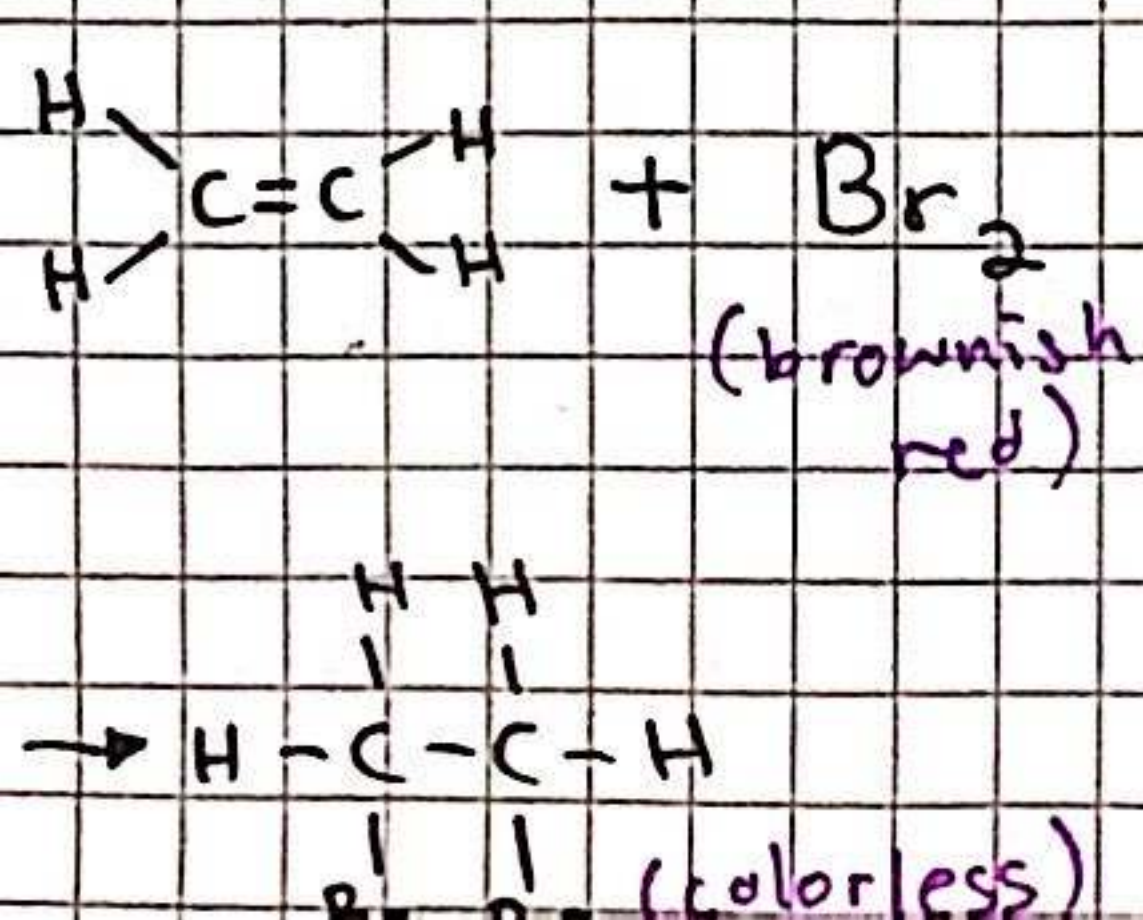
(obtained by fractional
distillation of petroleum)



(addition reaction)

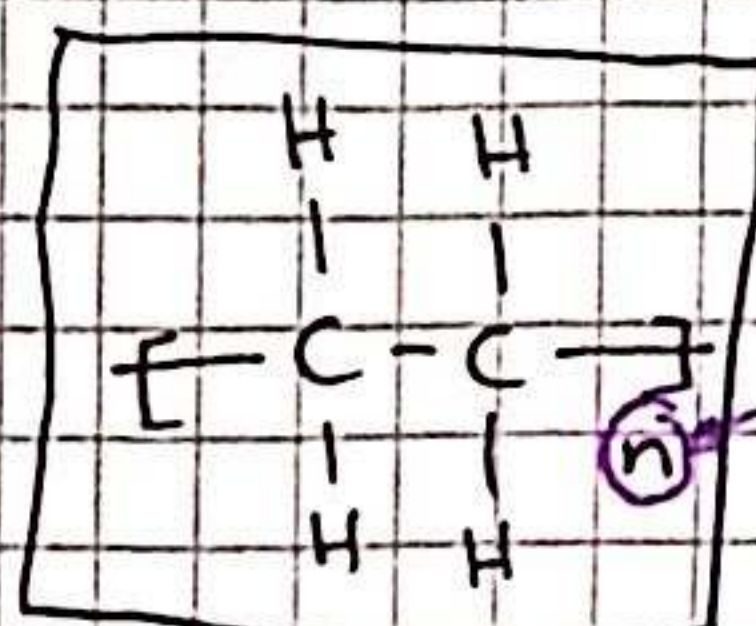
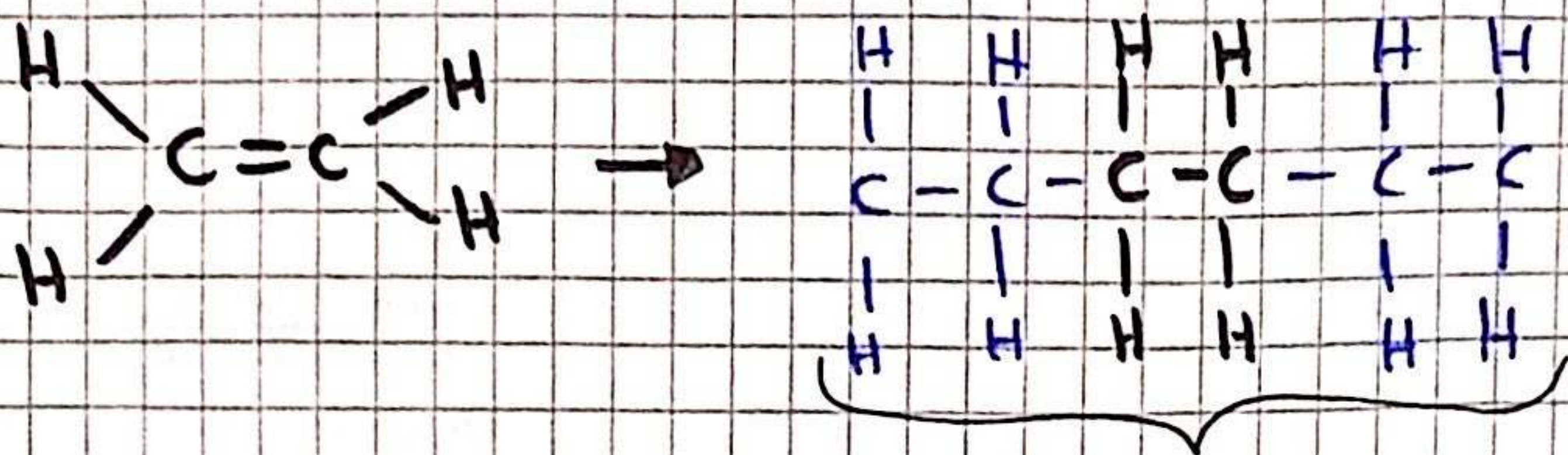
(i) the $C=C$ double bond can be broken to add
molecules onto the compound

(ii) can undergo addition reactions with bromine, hydrogen and steam.



(addition polymerization)

(i) polymerization is the formation of long chain molecules called polymers from a large number of smaller monomer molecules.



Shows that there is a chain of these molecules

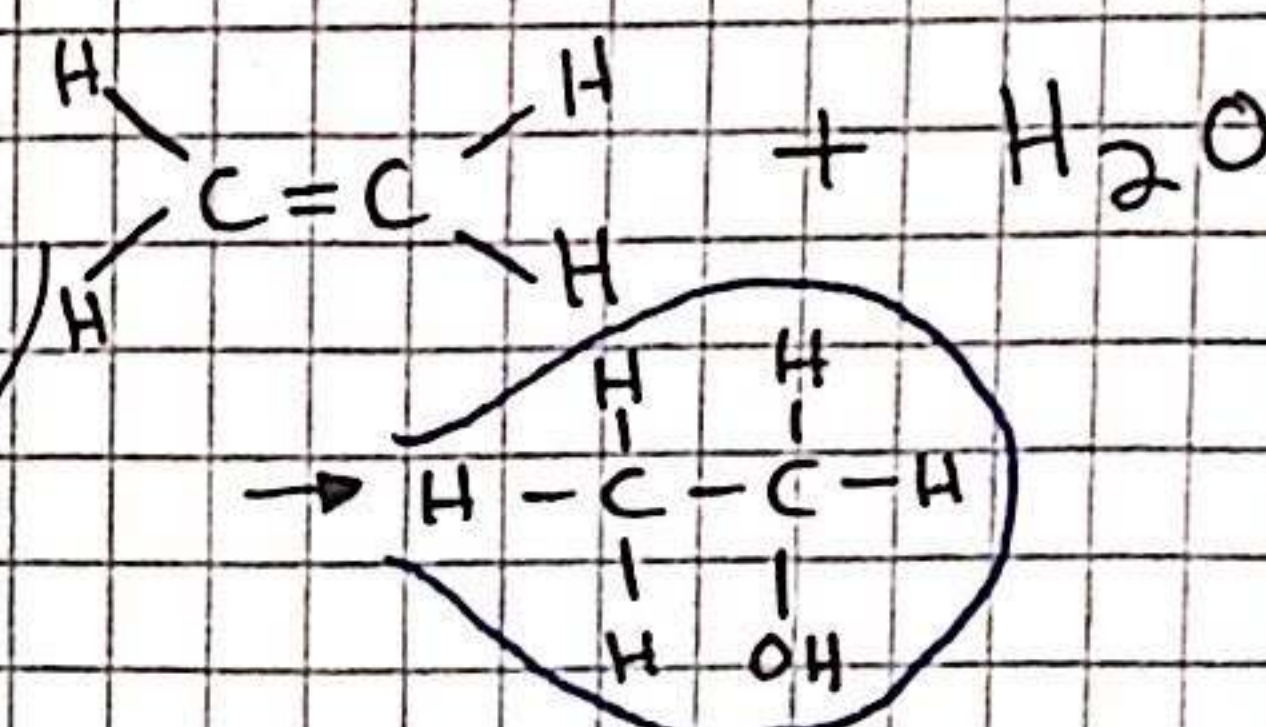
Alcohols

- general formula $\rightarrow C_nH_{2n+1}OH$
- ends with 'anol'

(ethanol manufacture)

(i) catalytic hydration of ethene

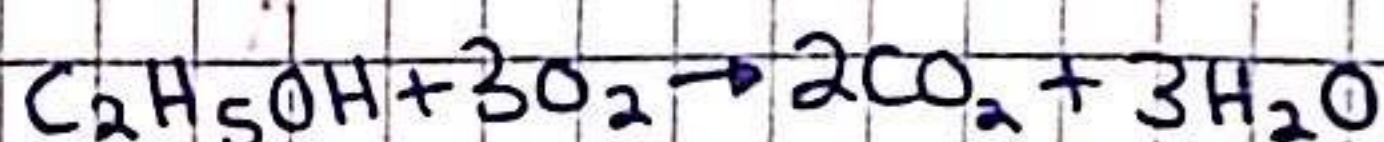
(ii) fermentation



- glucose $\xrightarrow{\text{yeast}}$ ethanol + CO_2
- the chemical breakdown of glucose by yeast
- CO_2 and ethanol is produced and the reaction is catalysed by yeast enzymes

ethanol:

- $\rightarrow$ burns with a blue flame
- $\rightarrow$ combustion of ethanol will produce carbon dioxide + H_2O



Catalytic hydration of ethene

advantages

- no waste product
- produces ethanol continuously

disadvantages

- use of crude oil (non-renew.)
- requires alot of energy

Fermentation

advantages

- + renewable (uses plants)
- + uses less energy

disadvantages

- slow, batch process
- produces CO_2

Ethanol can be used:

- as a fuel
- as a solvent in perfume and food industries
- in alcohol (some cultures)
- to make other organic materials such as ester.

$\uparrow$
needs new batch once yeast dies

Cracking

- breaks up large hydrocarbons into smaller and more useful bits

- needs high pressure and time w/o a catalyst and lower temperature and pressure with a catalyst.

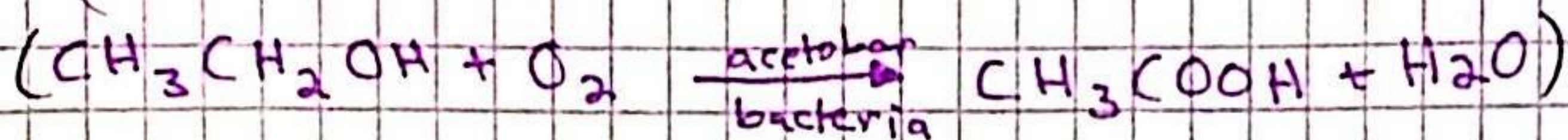
Carboxylic acids:

- general formula $\rightarrow C_n H_{2n+1} COOH$
- ends with '—oic acid'

(ethanoic acid manufacture)

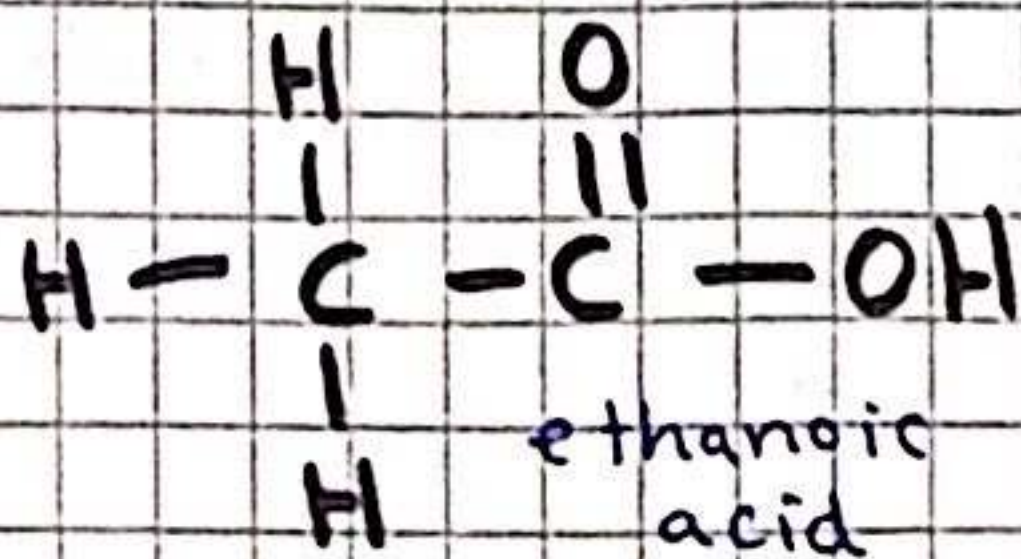
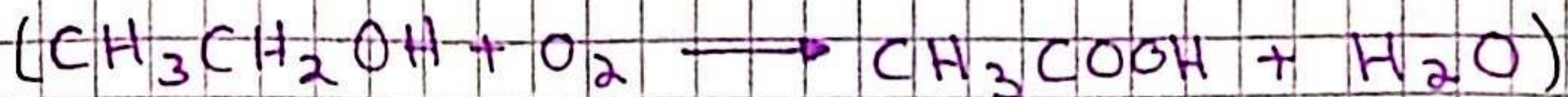
* (1) oxidation of ethanol via fermentation

- acetobacter bacteria can oxidise ethanol into ethanoic acid



* (2) oxidation of ethanol via acidified potassium manganate (VII)

- ethanol is heated with potassium manganate (oxidizing agent), ethanoic acid is formed



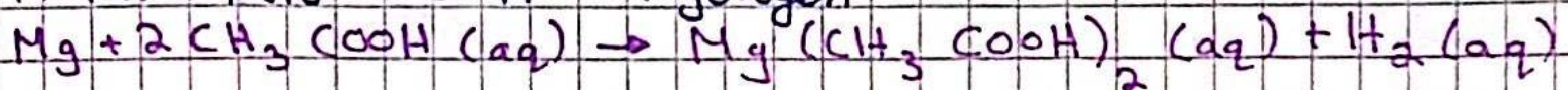
(ethanoic acid properties)

* all carboxylic acids are weak acids

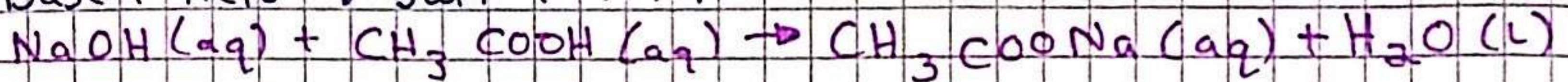
* they demonstrate typical acid properties and they only partially ionize in aqueous solution.

e.g.

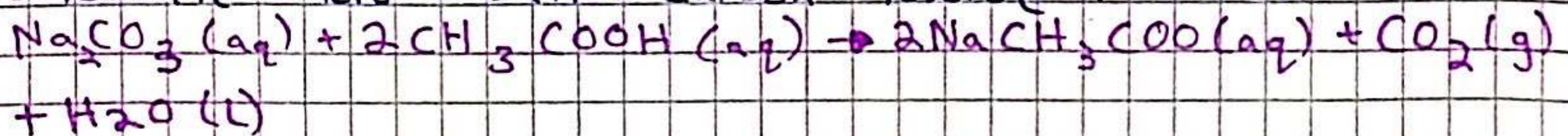
Metal + Acid $\rightarrow$ Salt + Hydrogen



Base + Acid $\rightarrow$ Salt + Water



Carbonate + Acid $\rightarrow$ Salt + Carbon dioxide + Water

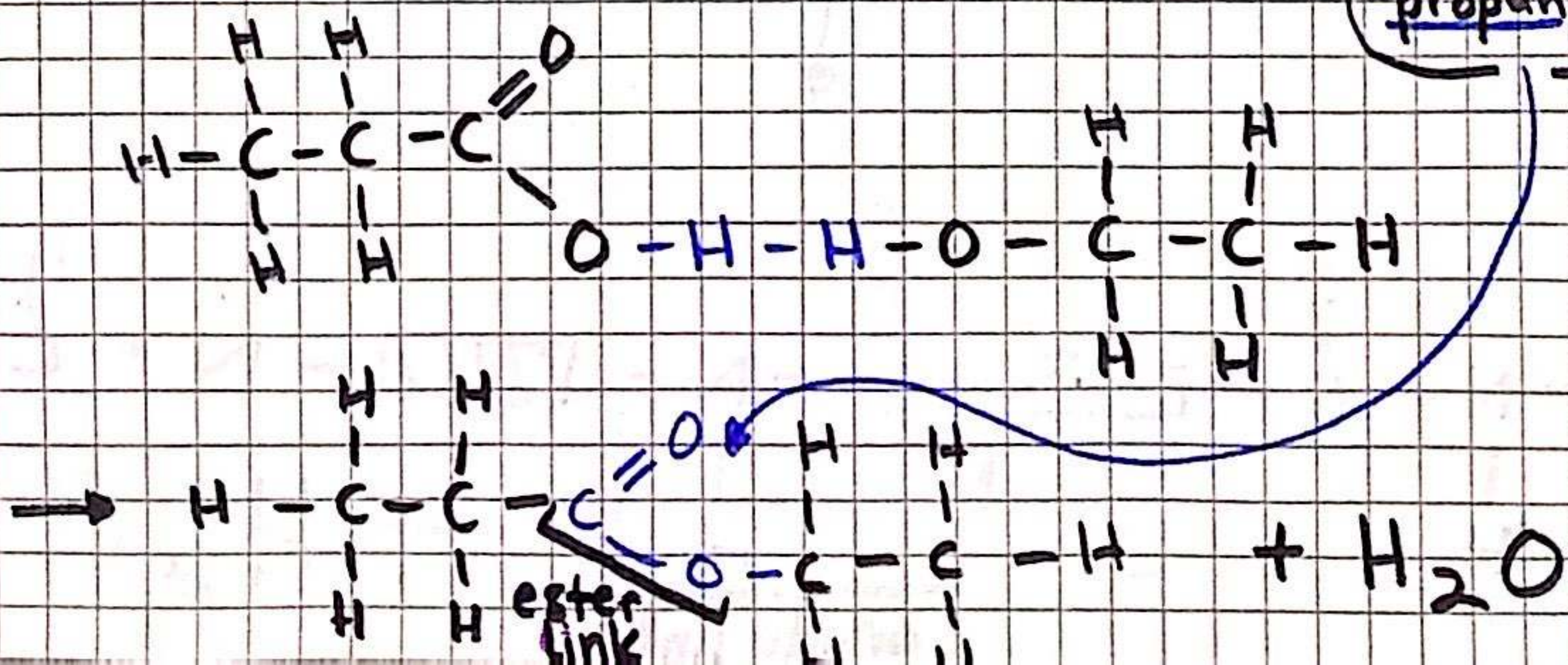


(esters)

• Carboxylic acids react with alcohol to form an ester + water

requirement:-

- heat
- concentrated sulfuric acid catalyst



ethyl propanoate

polymers:

- polymers are large molecules built from small units (monomers)
- different polymers are built different monomers and have varying linkages between the monomers

[two methods of polymerization]

- [addition]
- [condensation]

synthetic polymers

- [nylon]
- [terylene]

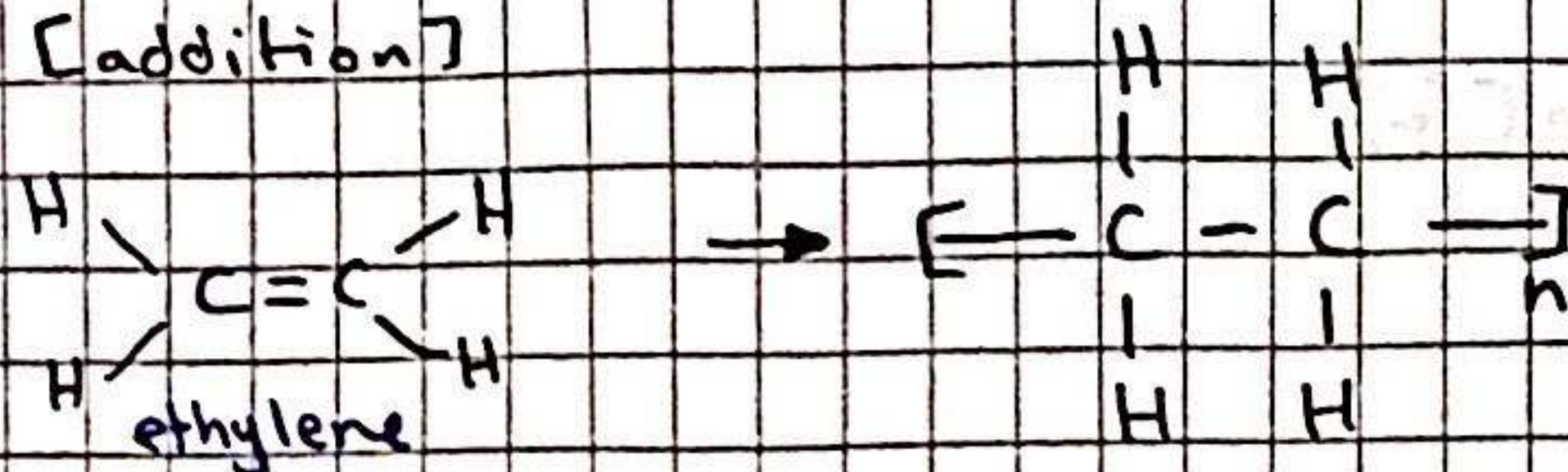
derived from petroleum oil,
made by scientists

natural polymers

- [protein]
- [carbohydrates]

occur in nature and
can be extracted

[addition]



- ethylene has 2 carbon and 4 hydrogen atoms, and the polyethylene repeat structure has 2 carbon and 4 hydrogen atoms. None gained, none lost.

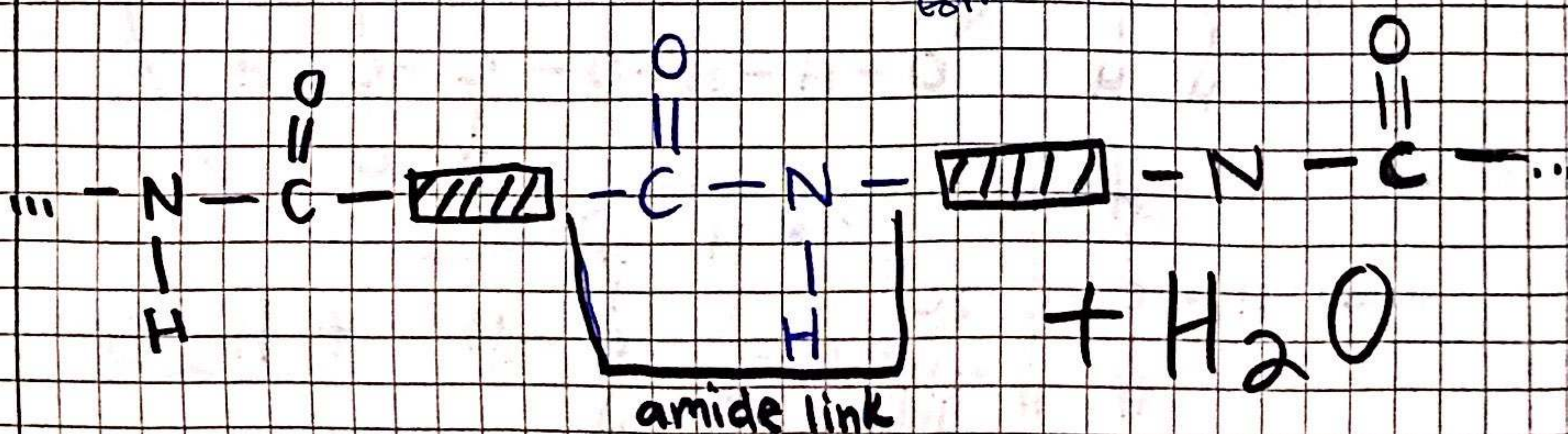
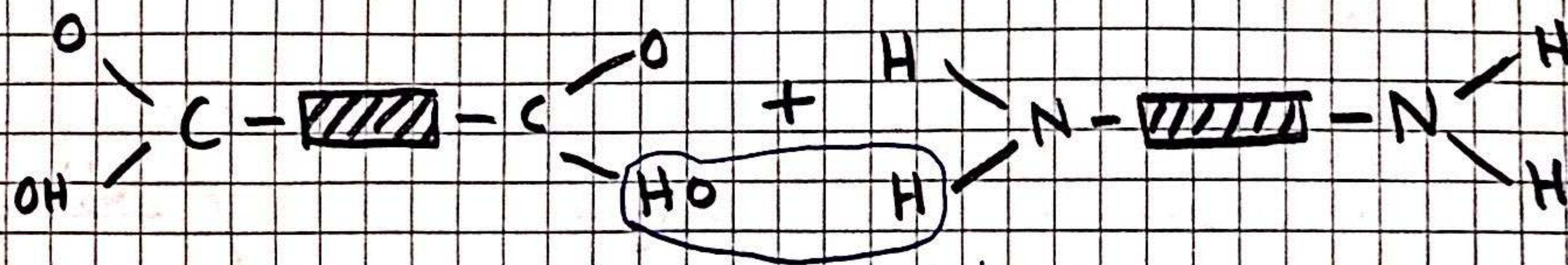
[condensation]

- two monomers react together and join. During this process a water molecule is lost (therefore condensation)
- the examples of synthetic and natural polymers above are made from condensation.

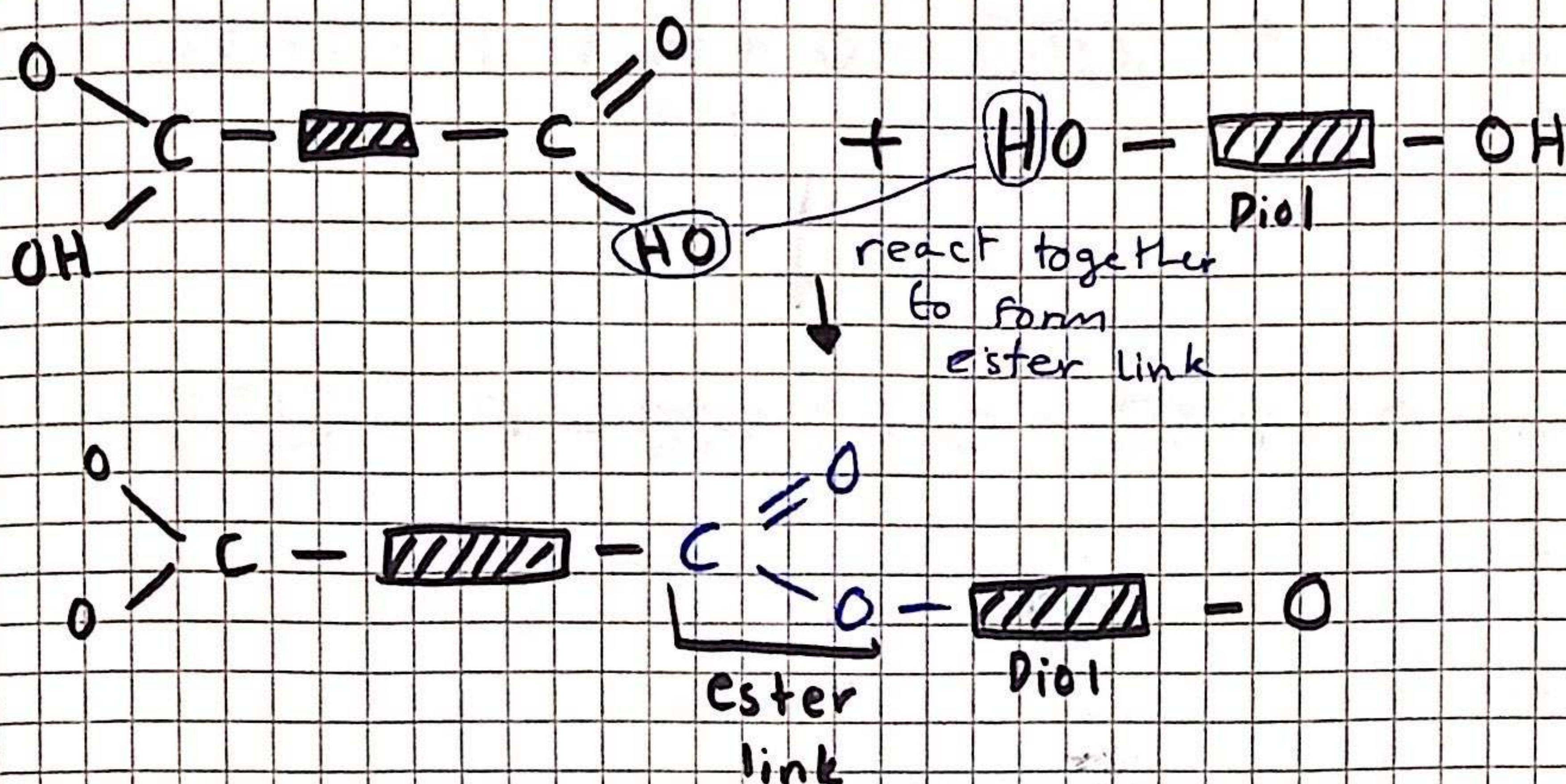
[synthetic polymer]

- nylon → polyamide
- two monomers link:

- dicarboxylic acid (diacid)
 - diamine
- forms an amide link

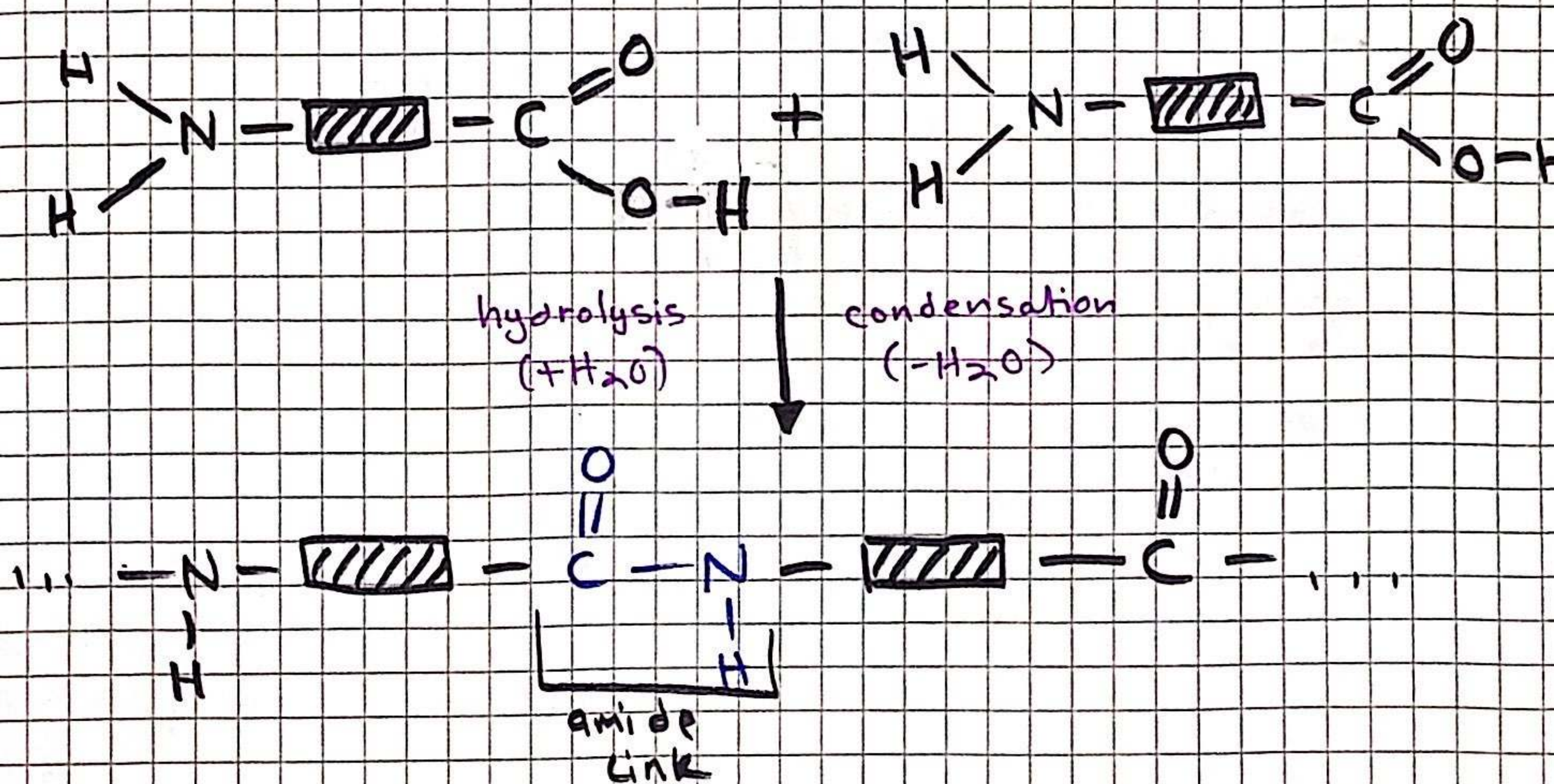


- terylene $\rightarrow$ polyester
- two monomers link:
 - dicarboxylic acid (diacid)
 - diol

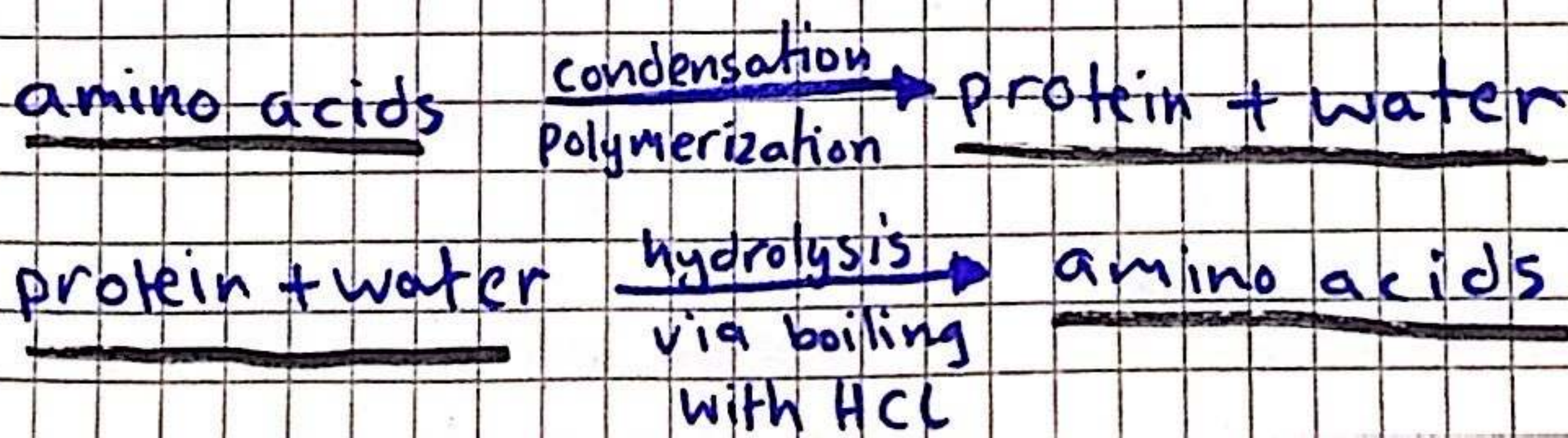


[natural polymers]

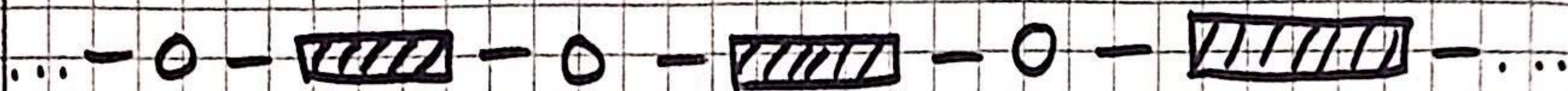
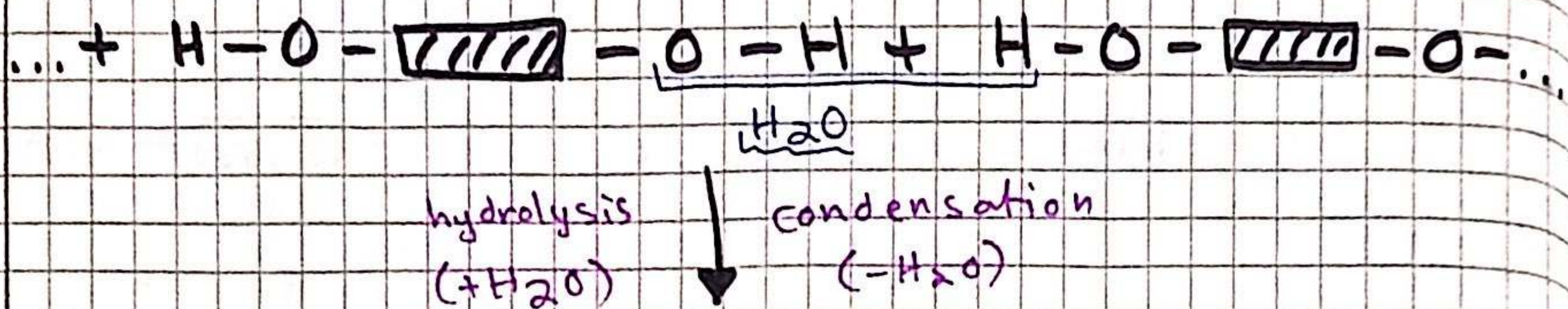
- protein $\rightarrow$ amino acids monomer
- amide link
- like nylon but different units



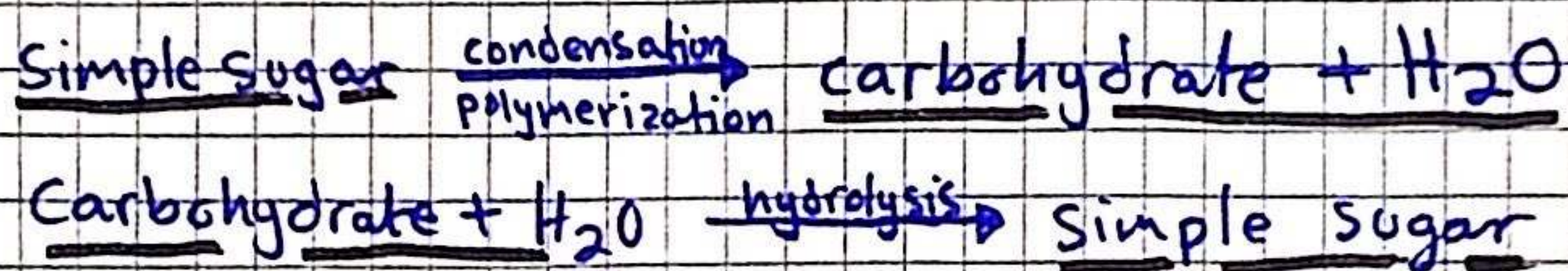
- amino acids can be joined via condensation polymerization to form protein
- proteins can then be hydrolyzed back to amino acids by boiling with hydrochloric acid (HCL)



- carbohydrates $\rightarrow$ simple sugar monomer
- made of simple sugar units



- Simple sugar monomers can be joined via condensation polymerization to form complex carbohydrates (ie starch)
- Complex carbohydrates can then be hydrolyzed back into simple sugar monomers.



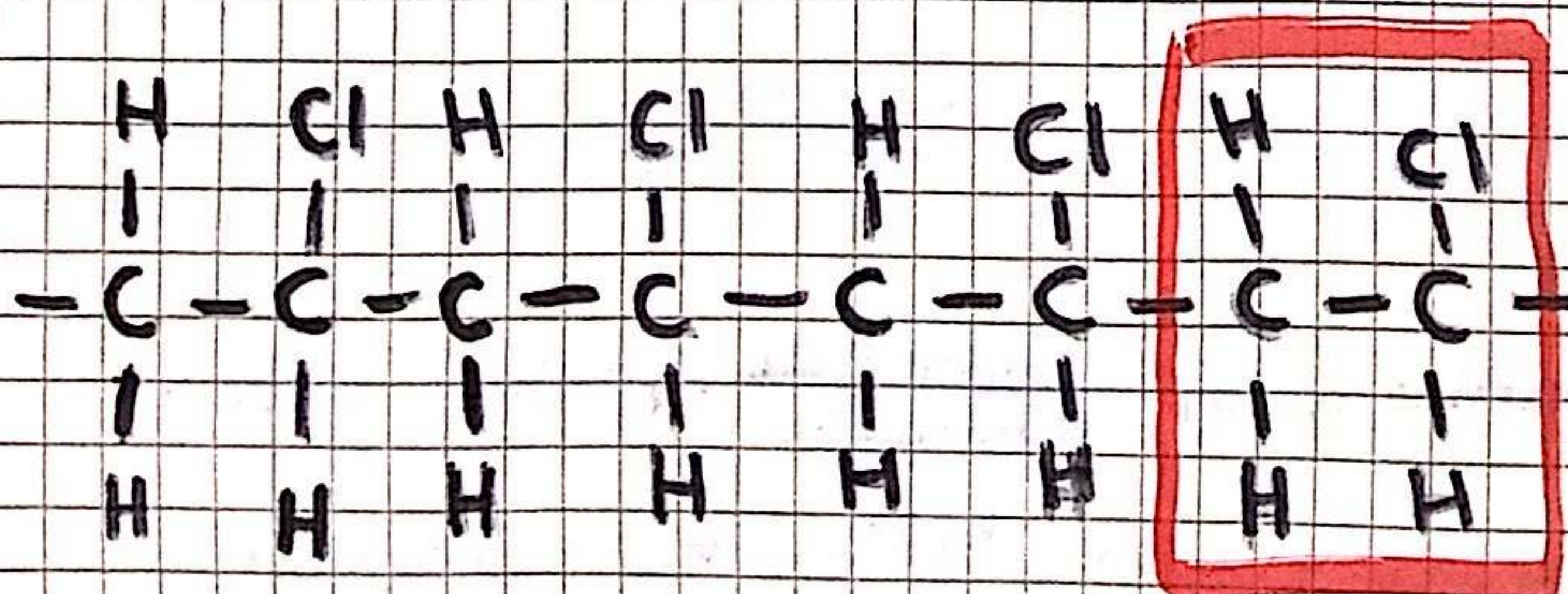
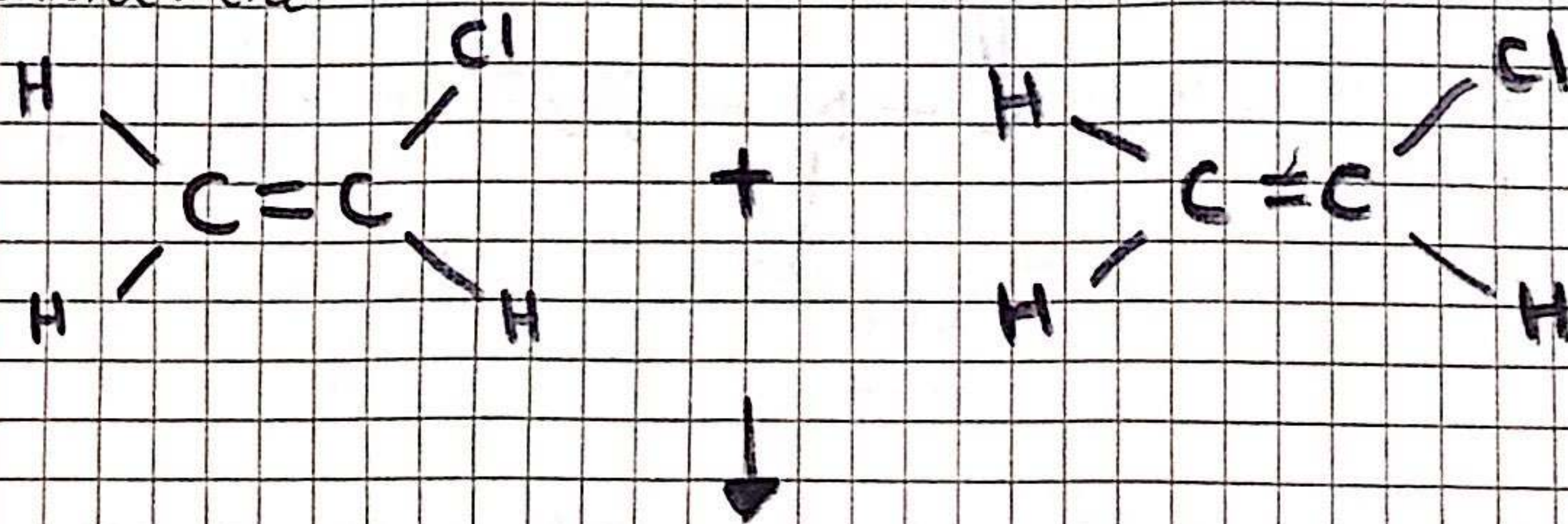
[Condensation polymerization]

- when smaller monomers react with each other to form large structural units such as polymers.

[Addition polymerization]

- when a chain reaction adds new monomer units to the growing polymer molecule one at a time through double or triple bonds.

- chloroethene



Fossil fuel:-

- a fuel is a substance, which, when burned, releases heat energy.
- → this heat can be transferred into electricity
- most common types of fossil fuels include; coal, natural gas and hydrocarbons such as methane and propane which are obtained from crude oil
- the main component of a natural gas is methane, CH_4

Petroleum:-

- petroleum (also known as crude oil) is a complex mixture of hydrocarbons which also contain natural gas.
- thick, sticky, black liquid found under porous rocks



crude oil

Sea bed

- petroleum itself as a mixture isn't very useful but each component part of the mixture, called a fraction, is useful and each fraction has different applications.
- Fractions in petroleum are separated from one another in a process called Fractional distillation.
- molecules in each fraction have similar properties and boiling points, which depend on the # of carbons in the chain.
- the boiling point and viscosity of each fraction increase as the carbon chain gets longer.

